28.03.2026 — 29.03.2026

HACKATHON CHALLENGE

Insurance market reverse engineering

The Scenario

You have just joined the data science team of a new insurer preparing to enter the car insurance market. To compete effectively, you need to understand how existing players price their policies without access to their internal models.

Your mission: **reverse-engineer the pricing strategies of real market competitors using only their quoted prices (prices = premiums) and the customer and vehicle data behind them.**

The Challenge

Part 1 Predictive Modeling

Build machine learning models that simulate each competitor's pricing logic. You will predict what price Insurers A through K would quote for any given row, which we call it insurance profile.

Your predictions will be evaluated on Mean Absolute Error (MAE), the closer your predicted prices are to the actual competitor quotes, the better.

Part 2 Strategic Insight

Numbers alone do not win markets. Beyond your models, you will need to tell a story:

- Which competitors are overpricing/underpricing certain customer segments?
- Who is targeting which drivers, high-value vehicles, or urban areas, and what segments are they avoiding?
- Where are the gaps in the market a new insurer could exploit?
- Come up and find your insights. This is an open question; everything you find in the data and think is meaningful you should showcase.

Your team will present a **market positioning recommendation**: how should your new insurer price itself to gain market share, and what would your pricing strategy look like.

The Data

You will receive **three data blocks**, all sourced from a real insurance comparison platform:

Block	Contents	Purpose
Block 1	Profiles + competitor prices (4 weeks)	Training data : build your models here
Block 2	Profiles, no prices (week 5)	Public leaderboard : submit predictions here
Block 3	Profiles, no prices (week 6)	Private leaderboard : used for final rankings

After submitting predictions for Block 2, you will receive **instant feedback** on your MAE score via the live leaderboard. Block 3 results are revealed only at the end of the hackathon.

Understanding the Dataset

One row = one insurance quote request. Each record represents a single profile visiting the comparison platform and requesting quotation for the policy. For that profile, you can see what price each competitor offered. Not every insurer quotes every profile, a missing price means the insurer declined to offer a quote for that particular combination of driver, vehicle and coverage. The features describing that profile (driver, vehicle, location) are the same across all competitors; only the prices and deductibles differ.

For each insurer, two columns are provided: a **price** column (the quoted premium) and a **deductible** column (the excess amount the customer would pay in case of a claim. This is insurer-specific, meaning different insurers may offer different deductibles for the same profile). In the training data (Block 1), both are included. In the test data (Blocks 2 and 3), the deductible columns are still present but the price columns are removed. And those remaining ones are what you need to predict. The deductible is a feature you may find useful when building your models. Please ignore missing prices, do not fill them with 0.

The data is real, and real data is messy. Expect missing values, edge cases and the occasional surprising entry. Handling this gracefully is part of the challenge.

The Features

Following sections describe key considerations when calculating a premium for a policy.

Coverage & Policy

The type of coverage requested: `mtpl` (third-party liability only), `limited_casco` (third-party + limited own damage), or `casco` (all-risk / full coverage), alongside the payment frequency and number of claim-free years.

A **claim** in motor insurance is when you ask your insurer to cover costs after an incident involving your car, such as an accident or damage. Depending on the type of insurance, the insurer may pay for your own damage (Casco) or for damage you caused to others (MTPL), after reviewing what happened.

A note on claim-free years: this value can be negative. In this Bonus Malus system, each newly reported claim in the policy results in a penalty of −5 claim-free years for the driver. A driver who has accumulated, say, 3 claim-free years and then reported a claim will drop to −2, yielding a negative value. The more negative the number, the worse the driver's claims history

Driver Profile

Date of birth of the primary contractor, whether the driver is also the vehicle owner, intended annual mileage, and usage type (private, business, etc.). Where a second driver is registered on the policy, their date of birth and claim-free years are also included.

Vehicle Characteristics

These features describe the physical and administrative properties of the insured car. Physical attributes include make, model, fuel type, engine size, power output, weight, dimensions (length, width, height), number of doors, seats, wheels and cylinders, primary color, and the estimated new value of the vehicle.

On the administrative side, you will find registration dates (first registration globally, first registration in the market), vehicle age, years since first registration in the given market, odometer readings and verdicts, and inspection report details including date, expiry date, and number of deficiencies found. Several binary flags are also included: whether the vehicle was imported, imported within the last 12 months, has an open recall, is marked for export, or is registered as a taxi.

Location & Regional Context

Each profile is linked to a 2-digit postal code region, an anonymized municipality, and an anonymized province. Beyond the raw location identifiers, the dataset includes a rich set of area-level statistics derived from national data. These capture the character of where the policyholder lives and can be powerful signals for risk:

- **Demographics**: population size, age group ratios, household composition (single-person, multi-person, with and without children), average household size
- **Housing**: urban category, average property value, and more
- **Crime & socioeconomics**: crimes per 1,000 inhabitants and more
- **Proximity to amenities**: distances to and counts of hospitals, schools, train stations and more
- **Geography**: distance to the national border, latitude, and longitude

These postal code features reflect the environment the vehicle is kept and driven in, which is something insurers weigh heavily when assessing risk.

Scoring

Submissions are evaluated using **Mean Absolute Error (MAE).** For each insurer k, the MAE is computed over the n_k rows where that insurer actually provided a quote:

$$\mathrm{MAE}_k = \frac{1}{n_k} \sum_{i=1}^{n_k} |y_{i,k} - \hat{y}_{i,k}|$$

Rows where the insurer did not quote (null prices) are excluded from scoring.

Your overall leaderboard score is a **single pooled MAE** across all eleven insurers. Accessing the leaderboard is done through this LINK. All absolute errors from all insurers are combined into one flat list and averaged:

$$\mathrm{MAE}_{\mathrm{overall}} = \frac{1}{\sum_k n_k} \sum_k \sum_{i=1}^{n_k} |y_{i,k} - \hat{y}_{i,k}|$$

Lower is better.

Final score will be determined based on two elements:

- Mean Absolute Erroe (MAE) – 60%
- Presentation to the jury – 40%

Point for the model precision will be awarded using the following approach. The lowest MAE will be awarded 60 points, with each subsequent rank receiving one point fewer (i.e., 60, 59, 58, …).

For pitching, each team will be evaluated based on the quality of their final presentation across the following criteria:

- Problem Understanding (max. 9 points)
- Presentation Quality & Teamwork (max. 9 points)
- Innovation & Creativity (max. 8 points)
- Methodology & Technical Approach (max. 7 points)
- Applicability & Feasibility (max. 7 points)

Teams will have 5 minutes for presenting their findings, for which they should prepare a short ppt presentation, followed by 2 minutes of questions from the jury.

Teams will be divided into two groups, with each group presenting in front of a designated jury panel. Each jury member will assign scores independently, and the final pitching score for each team will be calculated as the average of all jury members' scores.

The top two teams from each group will advance to a final round, where they will present again in front of the full jury. The final winner will be selected based on this final evaluation.

Submission format

Your prediction file should be a **csv** file with the following columns:

```
quote_id, Insurer_A_price, Insurer_B price, Insurer_C_price, Insurer_D_price,
    Insurer_E_price, Insurer_F_price, Insurer G price, Insurer H price,
        Insurer I price, Insurer J price, Insurer K price
```

Each row must correspond to a `quote_id` from the test block. You should submit a predicted price for every row and every insurer. The scoring system will automatically ignore predictions for rows where the insurer did not actually quote. Must use separator ";" and decimal "." with headers.

Deliverables

1. Prediction files for Block 2 (submitted during the hackathon) and Block 3 (submitted before the final deadline)
2. A short presentation covering your modeling approach and strategic market insights

Good luck, and may the best insurer win!